

FM3D-IAD: A Fine-Grained RGB–Point Cloud Dataset and Benchmark for Industrial Anomaly Detection

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Abstract—Reliable industrial inspection requires identifying subtle local defects while distinguishing them from valid variations in product appearance and geometry. Existing anomaly-detection benchmarks are dominated by RGB appearance or relatively regular normal patterns and therefore provide limited support for evaluating fine-grained RGB–point cloud (RGB-PC) inspection under heterogeneous multimodal evidence. We present FM3D-IAD, a dataset and benchmark comprising 2,492 aligned RGB–XYZ pairs from 10 industrial object categories, jointly acquired in each measurement using an industrial structured-light 3D camera. It covers heterogeneous materials, multiple production-acceptable normal variants, and controlled physical defects with small image-space support. For anomalous samples, pixel-level RGB masks and point-level point-cloud labels are manually provided according to defect visibility in each modality. These annotations support object-level detection, pixel-level RGB localization, and point-level 3D localization. A systematic benchmark evaluates representative RGB-only, point-cloud-only, and RGB-PC methods. Across anomalous test samples, 97.34% of RGB masks cover at most 0.5% of the full image area, while category- and sample-level statistics reveal heterogeneous defect scales and valid-point support in the point-cloud domain. We further provide GammaNet as a reproducible RGB-PC reference baseline for studying the complementary use of appearance and geometric information. The results reveal substantial category- and modality-dependent performance variation, establishing FM3D-IAD as a challenging testbed for fine-grained multimodal industrial anomaly detection. The dataset and code are available at <https://github.com/g77mak3r/FM3D-IAD>.

Index Terms—3D point clouds, fine-grained anomaly localization, industrial anomaly detection, multimodal inspection, RGB–point cloud data, structured-light imaging.

I. INTRODUCTION

INDUSTRIAL quality inspection requires detecting subtle defects while distinguishing them from acceptable variations in product appearance and geometry. Vision-based industrial anomaly detection (IAD) has advanced through reconstruction, knowledge distillation, feature embedding, memory-bank modeling, and application-specific defect detection [1], [2], [3], [4], [5], [6], [7]. However, precision-manufacturing defects are not limited to texture or color changes. They may involve local deformation, missing structures, or boundary damage that is difficult to characterize from RGB images alone. RGB–point cloud (RGB-PC) inspection addresses this limitation by combining high-resolution appearance with 3D geometric observations.

Recent datasets have extended IAD from 2D appearance in-

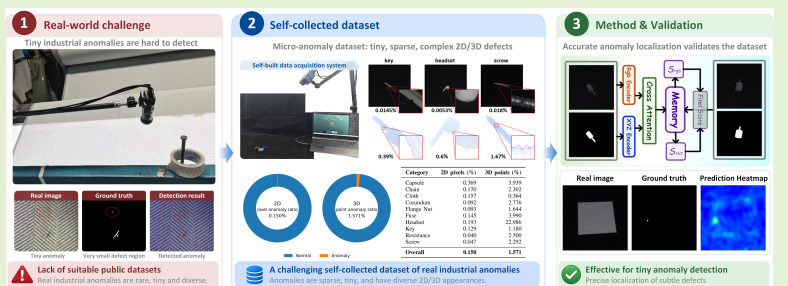
spection to more realistic or geometry-aware settings. MVTec 3D-AD and Eyecandies introduce RGB-3D multimodal data [8], [9], while Real3D-AD, Anomaly-ShapeNet, and MiniShift promote point-cloud anomaly detection under real, synthetic, or high-resolution 3D settings [10], [11], [12]. Real-IAD and 3CAD emphasize real production scenarios and large-scale high-resolution image inspection [13], [14], and CPS3D-Seg highlights the importance of point-level labels for fine-grained 3D defect characterization [15]. Related benchmark studies have paired RGB with hyperspectral data, combined measurements from multiple hyperspectral devices, or used structured-light point clouds for defect analysis [16], [17], [18]. Despite these advances, existing anomaly benchmarks are not primarily designed to evaluate fine-grained RGB-PC inspection under heterogeneous production-acceptable normal variation, as summarized in Table I.

As illustrated in Fig. 1, FM3D-IAD is motivated by three properties relevant to fine-grained industrial inspection. First, Fig. 1(a) illustrates **multi-variant normality**: samples assigned to the same normal category may exhibit visibly different acceptable shapes or appearances. In this paper, “normal variant” is used descriptively to denote such production-

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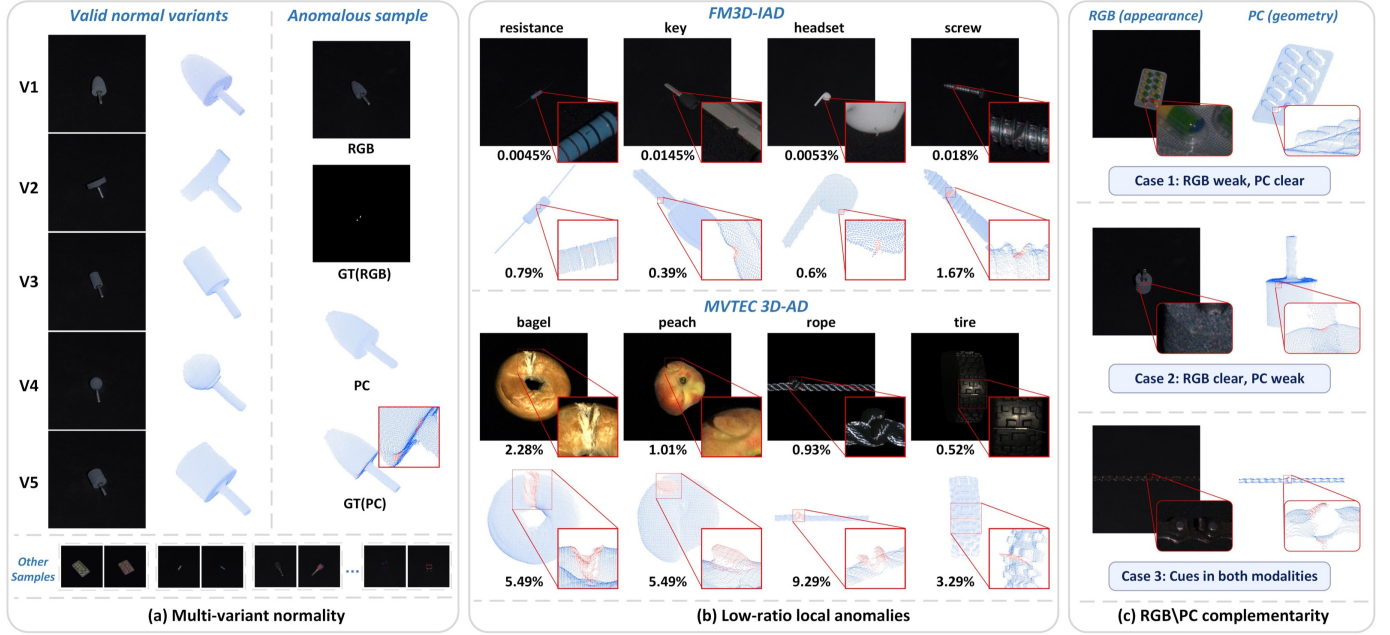


Fig. 1. Representative characteristics of FM3D-IAD. (a) **Multi-variant normality**: examples of visually or geometrically different normal samples that are all considered acceptable within the corresponding category. (b) **Low-ratio local anomalies**: representative MVTEc 3D-AD and FM3D-IAD samples are shown to illustrate differences in the spatial support of local defects; these examples are qualitative and are not intended as a distribution-level difficulty ranking. Quantitative anomaly-ratio statistics for FM3D-IAD are reported in Table VI and Figs. 5–6. (c) **RGB-PC complementarity**: representative cases in which anomaly evidence is more evident in RGB appearance, more evident in PC geometry, or distributed across both modalities.

TABLE I
ACQUISITION AND ANNOTATION CHARACTERISTICS OF REPRESENTATIVE INDUSTRIAL ANOMALY-DETECTION DATASETS.

Dimension	Attribute	MVTec AD [1]	LOCO-AD [25]	VisA [24]	MVTec 3D-AD [8]	Eyecandies [9]	Real-IAD [13]	FM3D-IAD (Ours)
Acquisition	Real-world data	✓	✓	✓	✓	×	✓	✓
	Paired RGB–3D data	×	×	×	✓	✓	×	✓
	Physically captured 3D data	×	×	×	✓	×	×	✓
Annotation	Pixel-level annotation	✓	✓	✓	✓	✓	✓	✓
	Point-level annotation	×	×	×	✓	✓	×	✓
Number of categories		15	5	12	10	10	30	10

acceptable intra-category variation. Second, Fig. 1(b) illustrates **low-ratio local anomalies**. The displayed cross-dataset examples are qualitative and are used only to provide visual context; quantitative claims are restricted to the dual-domain FM3D-IAD statistics reported in Table VI and Figs. 5–6. Third, Fig. 1(c) illustrates **RGB-PC complementarity**: different defects may be more evident in appearance, geometry, or both modalities. Together, these properties motivate evaluation under valid normal variation, small image-space anomaly support, and heterogeneous multimodal evidence.

To address these limitations, we introduce FM3D-IAD, an RGB-PC dataset and benchmark for fine-grained multimodal industrial anomaly detection. FM3D-IAD contains 2,492 aligned RGB image and XYZ point-cloud pairs across 10 industrial object categories. The pairs are acquired using an industrial structured-light 3D camera under fixed-view and controlled-illumination conditions. The dataset covers heterogeneous materials, production-acceptable normal variations, and controlled physical defects with small image-space support. For anomalous samples, pixel-level RGB masks and point-level point-cloud labels are manually annotated accord-

TABLE II
IMAGE-LEVEL AUROC OF REPRESENTATIVE METHODS ON DIFFERENT ANOMALY-DETECTION DATASETS.

Modality	Method	MVTec AD	Eyecandies	MVTec 3D-AD	FM3D-IAD
Single-RGB	DFM [21]	0.891	0.692	-	0.746
	STFPM [2]	0.876	0.708	0.793	0.634
	PaDiM [31]	0.950	0.794	0.764	0.707
RGB-D	M3DM [43]	-	0.897	0.945	0.637

ing to modality-specific visibility. These labels support object-level detection, pixel-level RGB localization, and point-level 3D localization.

We also provide GammaNet as a structured RGB-PC reference baseline. It extracts RGB appearance and point-cloud geometric representations, applies parameter-free cross-modal interaction, and constructs modality-specific memory banks for anomaly scoring. Table II provides cross-dataset reference results for several representative methods. Several methods obtain lower image-level AUROC on FM3D-IAD than on established benchmarks, although these values should be interpreted as contextual comparisons because the evaluation

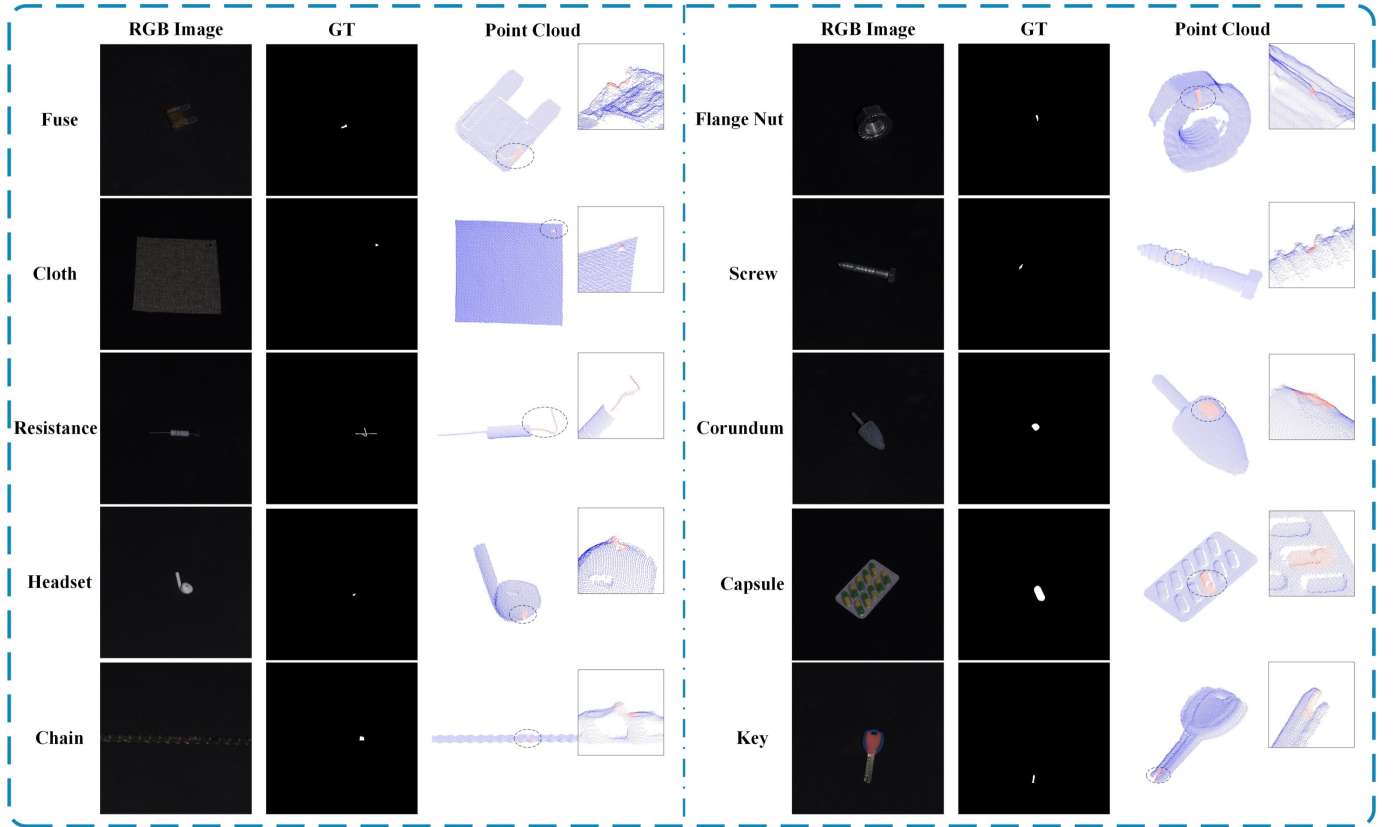


Fig. 2. Qualitative examples from FM3D-IAD across 10 object categories. From left to right, each panel shows the RGB image, the ground-truth (GT) pixel-level mask, and the corresponding XYZ point cloud. The circular indicators and zoomed-in insets highlight localized structural and geometric anomalies in the point clouds.

settings are not fully identical across datasets.

- We introduce FM3D-IAD, an RGB-PC dataset containing 2,492 aligned RGB image and XYZ point-cloud pairs from 10 industrial object categories. The dataset covers heterogeneous materials, production-acceptable normal variation, fine local structures, and controlled physical defects with small image-space support.
- We establish a controlled acquisition setup and dual-domain annotation procedure using an industrial structured-light 3D camera. Pixel-level RGB masks and point-level point-cloud labels are provided according to defect visibility in each modality, while dataset diagnostics characterize anomaly scales, heterogeneous valid-point counts, normal train-test point-count consistency, and potential point-count shortcut cues.
- We provide a systematic benchmark covering RGB-only, point-cloud-only, and RGB-PC methods for object-level detection, pixel-level RGB localization, and point-level 3D localization. GammaNet is included as a reproducible RGB-PC reference baseline.

II. RELATED WORK

A. Industrial Anomaly Detection Datasets

Industrial anomaly-detection datasets define both the learning setting and the evaluation protocol. MVTec AD established a representative unsupervised benchmark with diverse industrial objects and pixel-level masks [1]. Subsequent 2D datasets

TABLE III
STATISTICS AND INPUT MODALITIES OF REPRESENTATIVE ANOMALY-DETECTION DATASETS.

Modality	Dataset	Year	Type	#Class	Point Range
Single-RGB	MVTec AD [1]	2019	Real	15	-
	VisA [24]	2022	Real	12	-
	MVTec LOCO-AD [25]	2022	Real	5	-
	MPDD [26]	2021	Real	6	-
	BTAD [27]	2021	Real	3	-
	Real-IAD [13]	2024	Real	30	-
	MAD [28]	2023	Syn+Real	20	-
Single-Point	Real3D-AD [10]	2023	Real	12	35K–780K
	Anomaly-ShapeNet [11]	2024	Syn	40	8K–30K
RGB-Point	MVTec 3D-AD [8]	2022	Real	10	10K–30K
	Eyecandies [9]	2022	Syn	10	-
	FM3D-IAD (Ours)	2026	Real	10	1.84K–178.39K

expanded category scale, logical anomalies, real production conditions, pose variation, and high-resolution inspection, including VisA, MVTec LOCO AD, MPDD, BTAD, Real-IAD, MAD, and 3CAD [13], [14], [24], [25], [26], [27], [28]. However, these benchmarks primarily evaluate appearance cues.

Geometry-aware benchmarks extend the task beyond RGB. MVTec 3D-AD and Eyecandies provide RGB-3D or multimodal observations [8], [9]; Real3D-AD and Anomaly-ShapeNet focus on real or synthetic point clouds [10], [11]; MiniShift emphasizes high-resolution point clouds for subtle defects [12]; and CPS3D-Seg provides point-level labels for industrial 3D segmentation [15]. Table III summa-

TABLE IV
QUALITATIVE ANALYSIS OF APPEARANCE AND GEOMETRY CHALLENGES ACROSS THE TEN OBJECT CATEGORIES IN FM3D-IAD.

Category	Challenges in RGB Data	Challenges in XYZ Data
Capsule	Repeated capsules with similar appearance and tiny defect regions make subtle anomalies hard to observe under the dark background.	Repeated capsule-like geometry and shallow surfaces make local defects hard to separate from normal surface variations.
Chain	The chain is thin and elongated, with low foreground ratio, making small defects easily overwhelmed.	Repeated small links and sparse local points make point-level defect localization sensitive to sampling density.
Cloth	Dense woven textures make fine defects visually similar to normal fabric patterns.	Nearly planar geometry with height variations makes defects hard to distinguish from fabric unevenness.
Corundum	Dark appearance and irregular shape reduce RGB contrast, making small surface defects difficult to detect.	Irregular 3D geometry causes normal shape variations to be close to defect-induced deviations.
Flange Nut	The small dark metallic object and tiny defect area limit reliable RGB localization.	Circular holes, inner walls, and sharp edges make boundary defects easy to confuse with normal discontinuities.
Fuse	The compact fuse occupies a small image region, while its translucent casing and dark background reduce local contrast.	Thin terminals and a thin-walled housing lead to sparse point sampling, complicating the localization of small structural defects.
Headset	Small object scale and low contrast weaken visual cues for tiny local defects.	Compact curved parts and thin extensions require fine-grained geometric discrimination.
Key	Defects on the thin metallic blade occupy only a few pixels and are easily hidden by the small scale.	The thin blade and thicker handle introduce thickness ambiguity and sparse-point localization challenges.
Resistance	The tiny and slender object makes defects around leads or the body sensitive to resolution and illumination.	Slender leads and a small cylindrical body provide very few points for stable geometric anomaly scoring.
Screw	Repetitive thread patterns and shading variations can mask subtle RGB defects.	High-frequency helical geometry makes local defects difficult to separate from normal thread structures.

izes their input modalities and scales. FM3D-IAD complements these datasets by combining real RGB-PC acquisition, production-acceptable normal variation, low-ratio image-space defects, and modality-specific pixel/point annotations within one benchmark.

This distinction is important in precision manufacturing: a defect may preserve RGB appearance while changing local shape, or may alter appearance without producing an observable geometric response. A modality-aware benchmark must therefore retain both observations and provide separate localization labels.

B. Multimodal Benchmark Datasets and Imaging Systems

PCB-Vision pairs high-resolution RGB images with hyperspectral data cubes and provides masks and baseline models for printed-circuit-board analysis [16]. Koirala *et al.* acquired a hyperspectral benchmark with 13 devices and composition ground truth to study spectral unmixing under acquisition variability [17]. For geometric inspection, structured-light systems generate point clouds for 3D reconstruction and defect analysis [18]. These studies demonstrate how the acquisition setup, paired observations, annotations, and evaluation protocol jointly define a useful benchmark.

Image–point cloud methods also combine camera images with sparse geometric data for point-cloud enhancement or completion [19], [20]. Their designs highlight the importance of correspondence and feature alignment when appearance and

geometry are processed jointly. FM3D-IAD adopts the same practical organization: aligned RGB and XYZ observations, modality-specific annotations, and separate object-, pixel-, and point-level evaluation tasks.

C. Industrial Anomaly Detection Methods

Most unsupervised IAD methods model normal data through reconstruction, distillation, or feature embedding. DRAEM and RealNet use reconstruction or anomaly synthesis [29], [30]; STFPM and Reverse Distillation use teacher–student discrepancies [2], [4]; and PaDiM, PatchCore, CFA, and CFLOW-AD model pretrained feature distributions [3], [31], [32], [33]. Wave-Flow addresses unsupervised anomaly detection in photovoltaic electroluminescence images, while related industrial systems address component and laser-welding defect detection [5], [6], [7].

Together, these 2D families provide complementary baselines for appearance reconstruction, representation transfer, and normal-feature distribution modeling.

For 3D inspection, FPFH and BTF provide descriptor-based baselines [34], [35], while Point-BERT, PointMAE, R3D-AD, Reg3D-AD, ISMP, GLFM, PO3AD, and MC3D-AD develop learned geometric representations or reconstruction schemes [36], [37], [38], [10], [39], [40], [41], [42]. M3DM and multimodal reverse-distillation frameworks combine appearance and geometry [43], [44]. Camera–point-cloud fusion and cross-modal completion further show the importance of feature alignment and controlled information flow [19], [20].

TABLE V
DETAILED STATISTICS OF THE PROPOSED FM3D-IAD DATASET
ACROSS TEN OBJECT CATEGORIES.

Category	Anomaly Types	Material	Train	Test		Total
				Normal	Anomaly	
Capsule	4	Plastic, Gelatin	180	20	80	280
Chain	4	Metal	180	20	73	273
Cloth	6	Fiber	150	15	90	255
Corundum	4	Corundum	153	18	72	243
Flange Nut	4	Metal	86	18	78	182
Fuse	5	Plastic, Metal	180	20	100	300
Headset	5	Plastic, Metal	180	20	100	300
Key	4	Plastic, Metal	170	20	72	262
Resistance	4	Ceramic, Metal	135	21	61	217
Screw	5	Metal	106	11	63	180
Total	45	-	1520	183	789	2492

These considerations motivate a structured RGB-PC reference baseline for FM3D-IAD.

Pretrained point encoders provide local geometric descriptors, whereas reconstruction and prototype methods model deviations directly in point space. This distinction matters because strong RGB localization does not imply equally accurate 3D localization across heterogeneous categories.

III. FM3D-IAD DATASET AND ACQUISITION SETUP

A. Dataset Design and Inspection Scope

FM3D-IAD targets local industrial defects that must be distinguished from acceptable appearance and geometric variation using complementary RGB and XYZ information. Its 10 object categories span metal, plastic, corundum, fiber, ceramic, and gelatin and cover fine structures, reflective surfaces, deformable materials, repeated structures, and heterogeneous valid-point support. Table IV summarizes the category-specific appearance and geometry challenges.

Based on these object categories, we physically introduced 18 general anomaly patterns under controlled conditions, which were instantiated as 45 category-specific anomaly subtypes across the 10 categories. The controlled defects include micro-breakages, small holes, missing miniature components, scratches, and local structural deformations. Therefore, FM3D-IAD should be interpreted as a dataset of real observations with controlled physical defects, rather than a collection consisting exclusively of naturally occurring production defects. The dataset primarily targets small, localized defects, while some category-specific defects affect a larger portion of the observed 3D structure.

Fig. 2 provides representative RGB, pixel-level GT, and point-cloud examples from all ten categories, illustrating localized defects across diverse materials and structures and the complementary appearance and geometric evidence provided by the two modalities.

B. Multimodal Acquisition Setup

All RGB-PC pairs were acquired using a Zivid 2+ MR60 industrial structured-light 3D camera operated at a working distance of 600 mm. The device provides an RGB image and its corresponding XYZ point cloud during each acquisition. At the 600 mm working distance, the manufacturer-specified

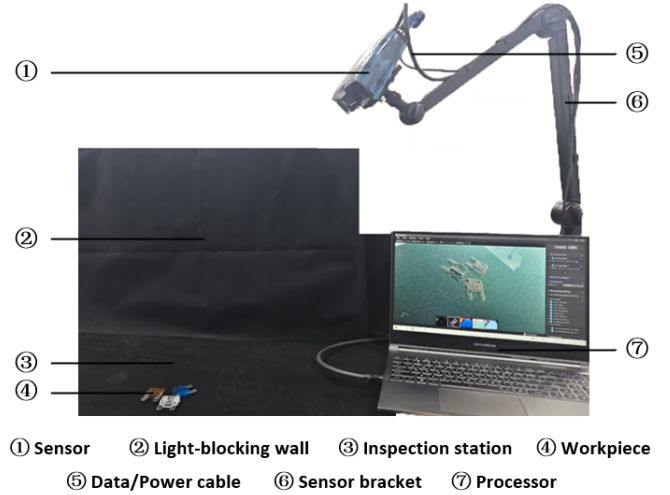


Fig. 3. Multimodal acquisition environment of FM3D-IAD.

nominal spatial resolution is approximately 0.24 mm and the typical point precision is 80 μm [45]. The native RGB resolution is 5.0 Megapixels (2448×2048 pixels), providing high-resolution appearance information with corresponding geometric observations.

For each acquisition, FM3D-IAD stores the paired RGB image and XYZ point cloud. The released point clouds retain XYZ coordinates without color attributes so that the 3D branch processes geometric information without duplicating RGB appearance. During data collection, the camera was statically mounted, samples were placed on a dark background, and indirect diffuse illumination was used. These controlled conditions reduce unrelated acquisition variations and preserve stable RGB-PC correspondence. A fixed rectangular crop was then consistently applied to the paired modalities, producing a unified 800×800 image-domain representation; the acquisition environment is shown in Fig. 3.

C. Dataset Statistics and Defect Distribution

As detailed in Table V, FM3D-IAD contains 1,520 normal training samples, 183 normal testing samples, and 789 anomalous testing samples, resulting in 2,492 aligned RGB-PC pairs. The dataset covers 10 industrial object categories and 45 category-specific anomaly subtypes. The split is designed to support unsupervised anomaly detection, where only normal samples are used for training and both normal and anomalous samples are used for testing.

Fig. 4(a) shows the category-wise numbers of normal training samples and anomalous test samples. Fig. 4(b) further decomposes the test set into normal samples (“Good”) and category-specific anomaly subtypes. The anomaly subtypes are instantiated from 18 general anomaly patterns and vary across object categories according to their physical structures and inspection requirements.

Each physical object was captured only once, and the resulting RGB-XYZ pair was assigned exclusively to either the training set or the test set. After acquisition, a normal specimen was removed from the collection process and was not reused to produce an anomalous sample.

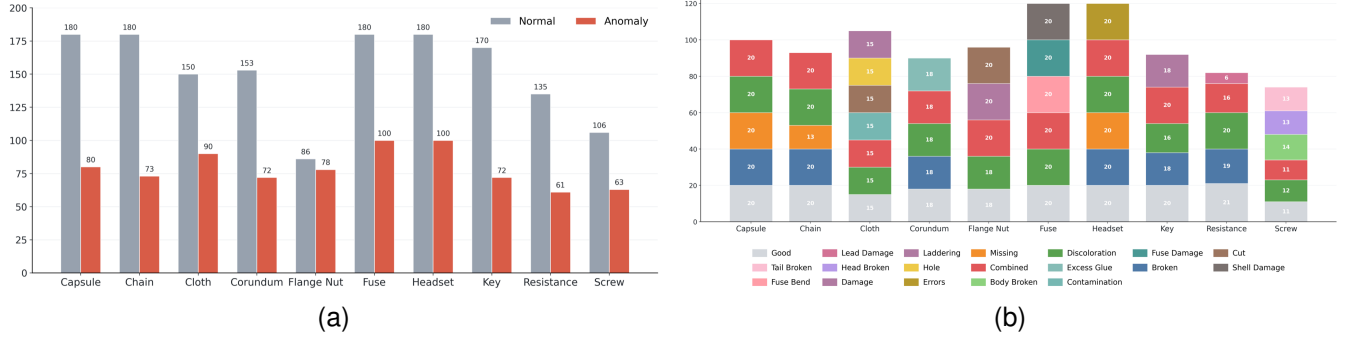


Fig. 4. Category-wise composition of FM3D-IAD. (a) Numbers of normal training samples and anomalous test samples. (b) Test-set composition, where “Good” denotes normal samples and the remaining segments denote category-specific anomaly subtypes.

TABLE VI

IMAGE-SPACE ANOMALY-AREA STATISTICS OF FM3D-IAD RELATIVE TO THE FULL IMAGE AREA.

Category	Masks	Median (%)	Q1–Q3 (%)	≤0.1%	≤0.5%
Capsule	80	0.418	0.265–0.460	23.75%	92.50%
Chain	73	0.073	0.048–0.158	63.01%	89.04%
Cloth	90	0.120	0.054–0.229	40.00%	96.67%
Corundum	72	0.044	0.027–0.097	75.00%	98.61%
Flange Nut	78	0.067	0.038–0.132	61.54%	100.00%
Fuse	100	0.128	0.054–0.242	44.00%	100.00%
Headset	100	0.112	0.018–0.394	48.00%	98.00%
Key	72	0.116	0.048–0.156	45.83%	98.61%
Resistance	61	0.025	0.015–0.053	90.16%	100.00%
Screw	63	0.035	0.023–0.060	90.48%	100.00%
Overall	789	0.078	0.034–0.208	55.77%	97.34%

TABLE VII

VALID-POINT STATISTICS OF THE FM3D-IAD POINT CLOUDS.

Category	Samples	Mean	Std.	Min	Max
Capsule	280	14,599	733	11,919	17,028
Chain	273	29,899	447	28,478	32,417
Cloth	255	170,146	3,137	162,762	178,385
Corundum	243	15,215	3,579	9,024	21,808
Flange Nut	182	25,288	1,396	15,970	27,216
Fuse	300	12,774	545	10,228	13,956
Headset	300	4,583	377	3,241	5,360
Key	262	17,552	1,708	14,299	21,001
Resistance	217	2,920	760	1,837	4,040
Screw	180	8,268	4,203	3,319	14,488
Overall	2492	30,443	47,875	1,837	178,385

D. Dual-Domain Defect-Scale Analysis

Table VI summarizes the category-wise and overall image-space anomaly statistics. The ratio is defined as the number of annotated anomalous pixels relative to the full image area. Under this definition, the median anomaly-region ratio across the 789 anomalous test samples is 0.078%, with an interquartile range of 0.034%–0.208%. Moreover, 55.77% of anomalous samples have annotated regions occupying no more than 0.1% of the image area, and 97.34% occupy no more than 0.5%. In the released images, the median anomalous region contains 500 pixels and the smallest annotated region contains 24 pixels. Resistance and Screw exhibit particularly small image-space anomaly regions, with median ratios of 0.025% and 0.035%, respectively.

For anomalous sample k , the modality-specific defect ratios are defined as

$$r_{2D}^{(k)} = \frac{N(M_k^{rgb})}{HW}, \quad r_{3D}^{(k)} = \frac{N(P_k^{def})}{N(P_k^{valid})}, \quad (1)$$

where $N(M_k^{rgb})$ is the number of positive pixels in the RGB mask, and $N(P_k^{def})$ and $N(P_k^{valid})$ are the annotated anomalous and valid XYZ points, respectively. These per-sample ratios are used for the category-wise distributions and sample traces in Figs. 5 and 6.

Fig. 5(a) shows that the 2D anomaly regions remain below 1% of the image area across the observed samples, although their distributions vary by category. In contrast, Fig. 5(b)

reveals a broader range of geometric anomaly support. Most categories remain concentrated at relatively low point ratios, while Headset’s higher anomaly-point ratios partly reflect its “miss” subtype, which involves large-scale structural loss. The per-sample traces in Fig. 6 further show pronounced within-category variation. Thus, FM3D-IAD contains consistently sparse 2D anomaly regions together with 3D anomalies spanning local to large structural changes; the two domains should not be interpreted as having identical anomaly scales.

E. Valid-Point Count Analysis

Valid-point counts are computed from all 2,492 released XYZ samples after excluding non-finite and zero-valued entries. As summarized in Table VII, they range from 1,837 to 178,385 per sample, with category means from 2.92K for Resistance to 170.15K for Cloth. Screw exhibits comparatively large intra-class variation (3.32K–14.49K). The overall standard deviation therefore mainly reflects cross-category variation in available geometric support rather than local acquisition noise.

F. Normal Train–Test Point-Count Consistency

Because the released point clouds retain heterogeneous native point counts, we compare the mean number of valid XYZ points between the *train/good* and *test/good* subsets

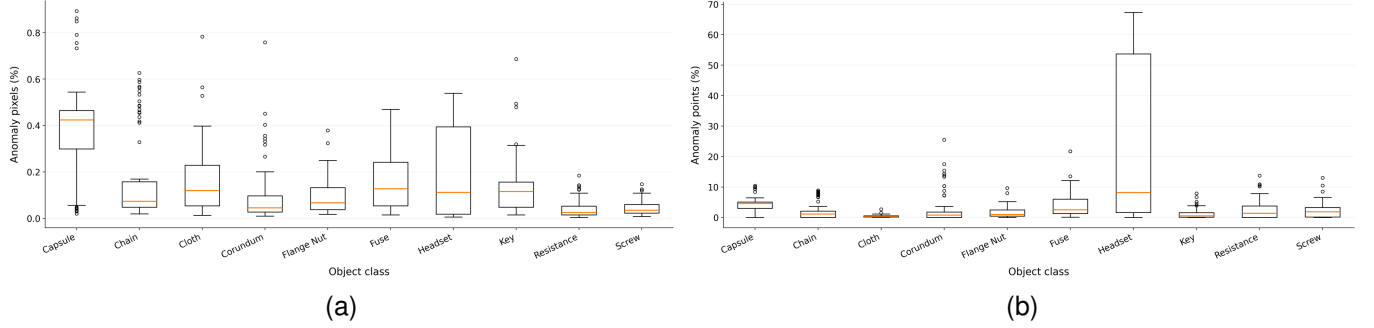


Fig. 5. Category-wise distributions of per-sample anomaly ratios. (a) 2D anomaly-pixel ratios. (b) 3D anomaly-point ratios. Center lines indicate medians, boxes span the first and third quartiles, whiskers show the non-outlier range, and circles denote outliers.

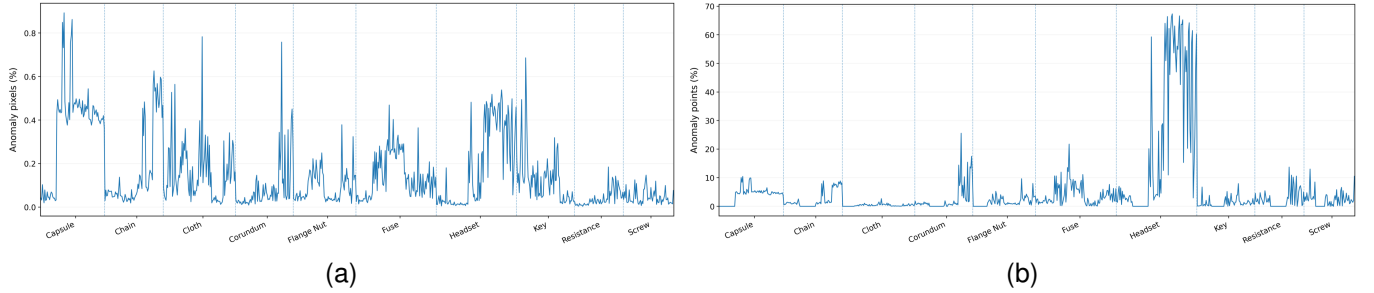


Fig. 6. Per-sample anomaly ratios grouped by object category. (a) 2D anomaly-pixel ratios. (b) 3D anomaly-point ratios. Vertical dashed lines separate categories; connecting lines are used only as visual guides and do not indicate temporal order.

TABLE VIII
POINT-COUNT CONSISTENCY AND SHORTCUT ANALYSIS ON FM3D-IAD.

Category	Train-Test Δ_N (%)	Point-Count Shortcut AUROC
Capsule	-3.60	0.544
Chain	-0.29	0.748
Cloth	+0.85	0.524
Corundum	-0.86	0.529
Flange Nut	+1.41	0.525
Fuse	+0.18	0.634
Headset	-0.18	0.538
Key	+1.45	0.536
Resistance	-1.33	0.653
Screw	-1.53	0.566
Mean	1.17	0.580

within each category. The relative difference is defined as

$$\Delta_N = \frac{\bar{N}_{\text{test}} - \bar{N}_{\text{train}}}{\bar{N}_{\text{train}}} \times 100\%.$$

As summarized in Table VIII, the absolute category-wise mean point-count difference remains below 4%, with an average absolute difference of 1.17%. Five categories differ by less than 1%, and the largest difference is observed for Capsule at -3.60% . These results indicate broadly similar valid-point-count statistics between the normal training and testing subsets.

G. Point-Count Shortcut Analysis

Because the released point clouds preserve heterogeneous native point counts, we further examine whether anomaly labels can be inferred from point-cloud size alone. For each category, the number of valid XYZ points is used as the sole anomaly score. To account for either increasing or decreasing point-count trends, we report

$$\text{AUC}_{\text{shortcut}} = \max(\text{AUC}, 1 - \text{AUC}),$$

where 0.5 corresponds to chance-level separability.

As shown in Table VIII, the mean of the ten category-wise shortcut AUROCs is 0.580. Nine of the ten categories remain below 0.70, while Chain reaches 0.748, indicating stronger category-specific separability from point count. On average, point count alone provides limited discrimination, although its predictive value varies across object categories.

H. Manual Dual-Domain Data Annotation

FM3D-IAD provides manual dual-domain annotations for all anomalous samples. All masks were manually produced and cross-checked by two authors. In the RGB domain, pixel-level masks were delineated using LabelMe according to visible defective regions. In the point-cloud domain, point-level labels were created using CloudCompare according to geometrically observable abnormal regions. The annotations are modality specific: a defect is marked in the RGB mask only where it is directly visible in RGB, and in the point-cloud mask only where it is geometrically observable in 3D. A defect region that is not observable in one modality remains zero in

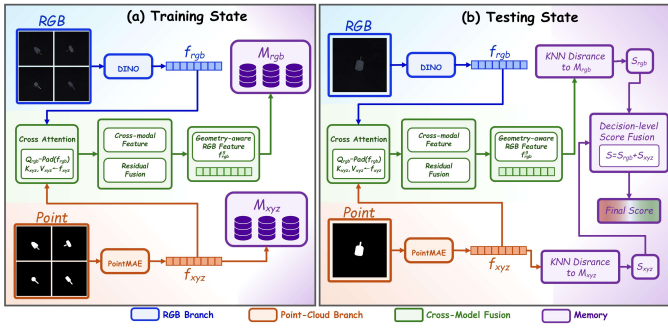


Fig. 7. Pipeline overview of GammaNet method.

that modality's mask; labels are neither copied nor projected from the other modality. Each released modality-specific mask is used directly as the ground truth for the corresponding localization evaluation.

IV. BASELINE MODEL: GAMMANET

To provide a reproducible reference for FM3D-IAD, we introduce **GammaNet**, a structured RGB-PC anomaly-detection baseline, as illustrated in Fig. 7. GammaNet follows the memory-based anomaly-detection paradigm of PatchCore [3] and consists of multimodal feature extraction, parameter-free cross-modal interaction, and modality-specific memory-bank scoring.

Multimodal feature extraction. Given an RGB image I^{rgb} and its corresponding point cloud P^{pc} , DINO [46] and PointMAE [37] extract the local token sets

$$\begin{aligned} F^{rgb} &= \Phi_{rgb}(I^{rgb}) \in \mathbb{R}^{N_{rgb} \times 768}, \\ F^{pc} &= \Phi_{pc}(P^{pc}) \in \mathbb{R}^{N_{pc} \times 1152}. \end{aligned} \quad (2)$$

Here, N_{rgb} and N_{pc} denote the numbers of RGB and PC tokens, respectively.

Parameter-free cross-modal interaction. GammaNet first extends each RGB token to the PC feature dimension by copying its first 384 channels:

$$Q = \mathcal{C}(F^{rgb}) = [F^{rgb}, F^{rgb}_{:,1:384}] \in \mathbb{R}^{N_{rgb} \times 1152}. \quad (3)$$

No additional ℓ_2 normalization or learnable projection is applied in this interaction module. The RGB-to-PC attention is computed by scaled dot products, with the softmax taken over all PC tokens for each RGB token:

$$A_{ij} = \frac{\exp(Q_i(F_j^{pc})^\top / \sqrt{1152})}{\sum_{\ell=1}^{N_{pc}} \exp(Q_i(F_\ell^{pc})^\top / \sqrt{1152})}. \quad (4)$$

The attended PC representation is $G = AF^{pc}$. Its first 768 channels are added to the original RGB tokens, whereas the PC tokens remain unchanged:

$$F^g = F^{rgb} + G_{:,1:768}, \quad F^p = F^{pc}. \quad (5)$$

Thus, the interaction is a one-way PC-to-RGB residual update, and F^g and F^p are subsequently scored in separate branches.

Dual memory-bank scoring. For $m \in \{rgb, pc\}$, let $H^{rgb} = F^g$ and $H^{pc} = F^p$. For each modality, μ_m and σ_m are scalar statistics jointly computed over all token positions

and all feature channels of all normal training tokens in that modality. All normal training-token entries are standardized as

$$\bar{H}^m = \frac{H^m - \mu_m}{\sigma_m}, \quad (6)$$

$$\mathcal{M}_m = \text{CoreSet}(\{\bar{H}_j^m\}_{\text{normal}}),$$

and the RGB and PC coresets are constructed independently. A test token is standardized by the corresponding training statistics and compared with its memory bank:

$$d_j^m = \min_{z \in \mathcal{M}_m} \|\bar{H}_j^{m, \text{test}} - z\|_2, \quad j_m^* = \arg \max_j d_j^m. \quad (7)$$

Let z_m^* be the nearest memory token to $\bar{H}_{j_m^*}^{m, \text{test}}$, and let $\mathcal{N}_2(z_m^*)$ contain the two nearest non-self memory neighbors of z_m^* . With D_m denoting the modality feature dimension, define

$$s_m^* = \frac{d_{j_m^*}^m}{1000}, \quad a_{m,z} = \frac{\|\bar{H}_{j_m^*}^{m, \text{test}} - z\|_2}{1000}, \quad z \in \mathcal{N}_2(z_m^*). \quad (8)$$

Here, 1000 is a fixed Euclidean-distance scaling factor shared by both modalities and kept unchanged across all object categories.

$$\begin{aligned} w_m &= 1 - \frac{\exp(s_m^* / \sqrt{D_m})}{\sum_{z \in \mathcal{N}_2(z_m^*)} \exp(a_{m,z} / \sqrt{D_m})}, \\ S_m &= w_m s_m^*. \end{aligned} \quad (9)$$

The final object-level score is the sum of the two modality-specific scores,

$$S = S_{rgb} + S_{pc}. \quad (10)$$

For localization, the RGB nearest-neighbor distances are reshaped to the token grid, interpolated to the input-image resolution, and smoothed to form the pixel-level anomaly map. The PC token distances are separately propagated to the original points by nearest-center assignment to form point-level anomaly scores; they are not merged with the RGB localization map.

Implementation Details. The RGB branch employs a DINO-pretrained ViT-B/8 initialized with the official weights released by Facebook Research. All RGB images are resized to 224×224 and normalized using the ImageNet statistics. We extract the patch tokens from the output of the final Transformer block after layer normalization while discarding the class token, yielding $28 \times 28 = 784$ patch representations of dimension 768. For the 3D branch, we employ a PointMAE-pretrained Point Transformer initialized from `pointmae_pretrain.pth`. The point encoder consists of 12 Transformer blocks with an embedding dimension of 384 and six attention heads. Each input point cloud retains all successfully parsed XYZ points and is partitioned into 1,024 local groups using farthest point sampling, with 128 nearest neighbors assigned to each group. The outputs of Transformer blocks 3, 7, and 11 are concatenated, resulting in a 1,152-dimensional representation for each point group. The RGB and point-cloud memory banks are independently compressed using a coreset retention ratio of 10% with sparse random projection ($\epsilon = 0.9$). The token-grid anomaly map is bilinearly resized to 224×224 and smoothed using a Gaussian filter with radius 4. For pixel-level evaluation, the resulting map is

TABLE IX

OBJECT-LEVEL ANOMALY DETECTION PERFORMANCE (AUROC) ON FM3D-IAD. BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY.

Method	Capsule	Chain	Cloth	Corundum	Flange Nut	Fuse	Headset	Key	Resistance	Screw	Mean
Single-RGB-based Anomaly Detection Methods											
PatchCore [3]	0.755	0.794	0.894	0.827	0.781	0.811	0.669	0.712	0.767	0.748	0.776
CFA [32]	0.727	0.849	0.888	0.817	0.631	0.812	0.783	0.736	0.611	0.723	0.758
CFLOW-AD [33]	0.576	0.696	0.555	0.322	0.598	0.749	0.637	0.530	0.690	0.865	0.622
DFM [21]	0.817	0.845	0.801	0.827	0.609	0.826	0.598	0.804	0.847	0.489	0.746
DFKDE [23]	0.668	0.843	0.579	0.662	0.534	0.552	0.738	0.499	0.696	0.550	0.632
PaDiM [31]	0.717	0.879	0.791	0.838	0.470	0.683	0.642	0.669	0.828	0.552	0.707
STFPM [2]	0.645	0.820	0.624	0.756	0.628	0.624	0.754	0.595	0.441	0.457	0.634
Single-Point Cloud-based Anomaly Detection Methods											
BTF (FPFH) [35]	0.481	0.582	0.526	0.353	0.487	0.578	0.699	0.524	0.685	0.365	0.528
BTF (Raw) [35]	0.579	0.828	0.645	0.559	0.444	0.750	0.580	0.498	0.559	0.518	0.596
PatchCore (FPFH) [3]	0.641	0.802	0.766	0.416	0.455	0.496	0.408	0.520	0.649	0.469	0.562
PatchCore (FPFH+Raw) [3]	0.643	0.933	0.683	0.487	0.444	0.595	0.371	0.466	0.582	0.704	0.591
PatchCore (PointMAE) [3]	0.568	0.780	0.526	0.534	0.526	0.526	0.441	0.495	0.679	0.620	0.570
M3DM (PointMAE) [43]	0.539	0.671	0.573	0.575	0.530	0.498	0.543	0.532	0.485	0.499	0.545
M3DM (PointBERT) [43]	0.546	0.746	0.520	0.596	0.454	0.577	0.564	0.594	0.457	0.680	0.573
Reg3D-AD [10]	0.553	0.827	0.539	0.520	0.328	0.554	0.544	0.481	0.633	0.541	0.552
RGB & Point Cloud-based Anomaly Detection Methods											
M3DM (PointMAE) [43]	0.579	0.925	0.864	0.594	0.474	0.554	0.537	0.497	0.528	0.499	0.605
M3DM (PointBERT) [43]	0.727	0.882	0.816	0.698	0.404	0.613	0.594	0.540	0.522	0.576	0.637
GammaNet	0.844	0.997	0.941	0.822	0.667	0.874	0.791	0.846	0.622	0.723	0.813

TABLE X

PIXEL-LEVEL AUROC ON FM3D-IAD IN THE RGB IMAGE DOMAIN.

Method	Capsule	Chain	Cloth	Corundum	Flange Nut	Fuse	Headset	Key	Resistance	Screw	Mean
Single-RGB-based Anomaly Detection Methods											
PatchCore [3]	0.982	0.987	0.946	0.916	0.975	0.993	0.998	0.989	0.974	0.983	0.974
CFA [32]	0.979	0.991	0.978	0.983	0.980	0.993	0.999	0.974	0.979	0.992	0.985
CFLOW-AD [33]	0.972	0.985	0.970	0.915	0.976	0.989	0.992	0.981	0.980	0.989	0.975
DFM [21]	0.941	0.741	0.849	0.798	0.943	0.956	0.966	0.931	0.921	0.679	0.873
PaDiM [31]	0.924	0.983	0.951	0.906	0.979	0.972	0.991	0.965	0.973	0.975	0.962
STFPM [2]	0.967	0.994	0.856	0.971	0.983	0.982	0.997	0.952	0.977	0.991	0.967
RGB & Point Cloud-based Anomaly Detection Methods											
M3DM (PointMAE) [43]	0.922	0.978	0.973	0.945	0.891	0.972	0.992	0.934	0.945	0.979	0.953
GammaNet	0.986	0.992	0.989	0.979	0.989	0.994	0.998	0.988	0.965	0.981	0.986

bilinearly upsampled to 800×800 and compared directly with the released RGB ground-truth mask. All experiments were conducted on an NVIDIA GeForce RTX 3090 GPU.

Baseline Implementations. All RGB-only (2D) baselines were implemented using Anomalib [22] with its default or recommended configurations. The point-cloud-only (3D) and existing RGB-PC baselines were reproduced from their corresponding official code repositories using the authors' original configurations unless otherwise stated. All methods were evaluated using the same FM3D-IAD training and test splits.

V. BENCHMARKS

To evaluate current methods on FM3D-IAD, we benchmark representative RGB-only, point-cloud-only, and RGB-PC approaches under the same dataset split. As summarized in Tables IX–XII, the benchmark covers object-level anomaly detection, pixel-level localization in the RGB image domain, and point-level localization in the 3D point-cloud domain.

Evaluation metrics. We use the area under the receiver operating characteristic curve (AUROC) for object-level anomaly detection and employ AUROC, maximum F1 score (F1-max), and area under the precision–recall curve (AUPR) for pixel- and point-level anomaly localization. Mean is the unweighted average across the ten categories. RGB-only and RGB-PC methods with image-space outputs are evaluated against the manually annotated RGB masks, whereas methods with point-space outputs are evaluated directly against the manually annotated point-level masks. Red and blue indicate the best and second-best results, respectively.

A. Object-Level Anomaly Detection

Table IX reports object-level AUROC for the three method groups. GammaNet achieves the highest mean AUROC of 0.813, exceeding the strongest RGB-only baseline, PatchCore, by 0.037 and the strongest existing RGB-PC baseline, M3DM (PointBERT), by 0.176. Its category-wise scores are particularly high for Chain (0.997), Cloth (0.941), Fuse (0.874),

TABLE XI
POINT-LEVEL AUROC ON FM3D-IAD IN THE 3D POINT-CLOUD DOMAIN.

Method	Capsule	Chain	Cloth	Corundum	Flange Nut	Fuse	Headset	Key	Resistance	Screw	Mean
Single-Point Cloud-based Anomaly Detection Methods											
BTF (FPFH) [35]	0.532	0.594	0.705	0.616	0.490	0.533	0.827	0.560	0.501	0.480	0.584
BTF (Raw) [35]	0.514	0.621	0.722	0.428	0.571	0.637	0.480	0.563	0.482	0.565	0.558
PatchCore (FPFH) [3]	0.551	0.560	0.606	0.548	0.488	0.474	0.744	0.564	0.567	0.459	0.556
PatchCore (FPFH+Raw)	0.548	0.630	0.680	0.551	0.478	0.517	0.738	0.582	0.483	0.462	0.567
PatchCore (PointMAE) [3]	0.521	0.417	0.515	0.519	0.456	0.441	0.526	0.494	0.576	0.467	0.493
M3DM (PointMAE) [43]	0.618	0.560	0.689	0.516	0.528	0.491	0.624	0.487	0.872	0.263	0.565
M3DM (PointBERT) [43]	0.597	0.543	0.637	0.482	0.516	0.483	0.630	0.461	0.623	0.603	0.558
Reg3D-AD [10]	0.488	0.477	0.612	0.577	0.481	0.420	0.507	0.478	0.579	0.541	0.516
GammaNet	0.554	0.523	0.629	0.518	0.566	0.434	0.552	0.411	0.481	0.417	0.509

TABLE XII
PIXEL- AND POINT-LEVEL F1-MAX AND AUPR OF GAMMANET ON FM3D-IAD.

Metric	Capsule	Chain	Cloth	Corundum	Flange Nut	Fuse	Headset	Key	Resistance	Screw	Mean
Pixel-F1-max	0.309	0.277	0.293	0.211	0.217	0.308	0.727	0.226	0.162	0.161	0.289
Pixel-AUPR	0.222	0.184	0.232	0.117	0.126	0.213	0.772	0.123	0.062	0.078	0.213
Point-F1-max	0.073	0.056	0.032	0.056	0.050	0.064	0.330	0.018	0.039	0.038	0.076
Point-AUPR	0.037	0.024	0.008	0.025	0.018	0.028	0.221	0.007	0.021	0.015	0.040

and Key (0.846), while Flange Nut, Resistance, and Screw remain comparatively more difficult. Among the single-modality baselines, PatchCore and CFA obtain mean AUROCs of 0.776 and 0.758 in the RGB domain, whereas BTF (Raw) gives the strongest point-cloud-only mean of 0.596. The different category rankings across method groups are consistent with the heterogeneous appearance, geometry, and defect characteristics summarized in Table IV.

B. Pixel-Level Anomaly Localization

Table X reports pixel-level AUROC in the RGB image domain. GammaNet achieves the highest mean AUROC of 0.986 under joint RGB-PC input, followed closely by CFA at 0.985. GammaNet reaches at least 0.979 in nine categories and obtains 0.994 on Fuse, 0.989 on Cloth and Flange Nut, and 0.998 on Headset; its lowest category score is 0.965 on Resistance. RGB-only methods also provide strong pixel-ranking performance, but their relative ordering changes across categories, reflecting differences in object scale, texture, and defect appearance. Overall, the high AUROC values indicate that anomalous pixels generally receive higher scores than normal pixels, although AUROC alone does not fully characterize the quality of a thresholded anomaly region under the highly imbalanced pixel distributions of FM3D-IAD.

C. Point-Level Anomaly Localization

Table XI reports point-level AUROC for point-cloud-only methods and GammaNet. The anomaly score assigned to each valid 3D point is compared directly with its manually annotated point-level ground-truth label, without projection to the RGB image plane. BTF (FPFH) obtains the highest mean AUROC of 0.584, followed by PatchCore (FPFH+Raw) at 0.567 and M3DM (PointMAE) at 0.565, while GammaNet obtains 0.509 from its modality-specific PC token distances.

The category-wise best results are distributed among different methods: BTF (FPFH) reaches 0.827 on Headset, M3DM (PointMAE) reaches 0.872 on Resistance, and BTF (Raw) reaches 0.722 on Cloth and 0.637 on Fuse. This variation indicates that no single geometric representation is uniformly strongest across the ten categories and that point-level localization is sensitive to object geometry, valid-point support, and defect scale.

D. Complementary Localization Metrics

Table XII complements the AUROC results with GammaNet's pixel- and point-level F1-max and AUPR. The mean pixel-level F1-max and AUPR are 0.289 and 0.213, respectively. Headset obtains the highest values of 0.727 and 0.772, whereas most other categories have pixel-level F1-max values between 0.161 and 0.309, reflecting the difficulty of delineating very small anomalous regions. In the point-cloud domain, the mean F1-max and AUPR decrease to 0.076 and 0.040; Headset again gives the highest category values of 0.330 and 0.221. The gap between AUROC and the F1/AUPR measures is consistent with the strong normal-anomalous imbalance in both domains and shows that ranking anomalous elements and producing precise localized regions remain distinct challenges.

E. Qualitative Results

Fig. 8 presents one anomalous RGB sample and its GammaNet response map for each category. The inset shows the corresponding pixel-level ground-truth mask, and warmer colors indicate higher anomaly scores. Across categories, the response maps place elevated scores around the annotated defect regions despite substantial differences in object size, shape, and appearance. Compact responses can be observed for small components such as Resistance and Fuse, while elongated or repeated structures such as Chain, Key, and

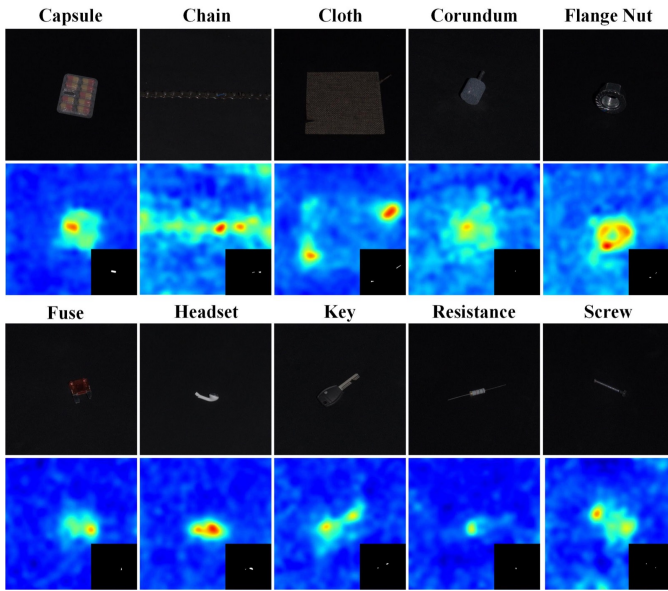


Fig. 8. Qualitative anomaly localization results on FM3D-IAD. Within each category panel, an anomalous RGB sample is shown above its corresponding GammaNet anomaly response map. The inset displays the pixel-level ground-truth mask, and warmer colors denote higher anomaly scores.

Screw produce responses with different spatial extents. These examples provide a visual complement to the quantitative results by showing how the predicted anomaly evidence is distributed around fine local defects.

VI. CONCLUSION

We presented FM3D-IAD, an RGB-PC dataset and benchmark comprising 2,492 aligned pairs across ten industrial categories, with pixel- and point-level annotations.

Dataset diagnostics show that most image-space anomalies are small: the median ratio is 0.078%, and 97.34% occupy no more than 0.5% of the image. The 3D anomaly scales and valid-point counts vary across categories, and the category-wise distributions retain both local defects and larger structural changes rather than reducing 3D difficulty to a single anomaly scale. Point count alone provides limited mean label separability (shortcut AUROC 0.580), although Chain shows a stronger category-specific correlation.

Benchmarks for RGB-only, point-cloud-only, and RGB-PC methods reveal substantial variation across categories and modalities. GammaNet serves as a reproducible RGB-PC reference baseline. FM3D-IAD therefore supports controlled study of appearance–geometry complementarity under acceptable normal variation and low-ratio image-space defects.

VII. LIMITATIONS AND FUTURE WORK

FM3D-IAD uses fixed-view, controlled-illumination acquisition to stabilize structured-light imaging and RGB-PC correspondence. It does not cover large viewpoint changes, uncontrolled illumination, or complex production backgrounds; future versions will extend the acquisition conditions while preserving cross-modal correspondence.

The current release includes ten precision-manufacturing categories with diverse materials, valid normal variants, and defect patterns. Expanding the category scale and collecting more naturally occurring defects would improve coverage of manufacturing processes. Future work will examine how point-density degradation, illumination variation, and other acquisition artifacts affect 3D localization. Stronger point-cloud foundation models and adaptive cross-modal interaction remain directions for future study.

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